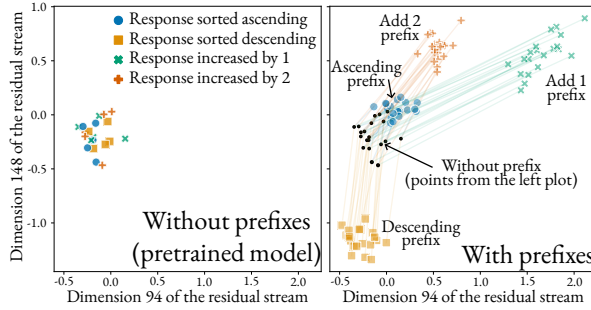


Figure 3: Attention block activations for ten sequences at the last input position (10) when pretrained on the four tasks. The left plot shows the pretrained activations t_{10} are not predictive of the completion. The right plot shows prefixes cluster the activations t_{10}^{pt} . Connecting the pretrained and prefixed activations highlights the bias. No dimensionality reduction is used; the clustering is solely due to the prefixes.



but can bring to the surface *latent knowledge* present in the pretrained model.⁷ We test this hypothesis by constructing small transformers trained on one or few tasks. We use a minimal transformer model (Karpathy, 2020) to show that prefix-tuning struggles to learn a new task that full fine-tuning can. Then, that prefix-tuning can easily elicit a latent skill from pretraining. Finally, we show how it can even learn *some* new tasks, provided they can be solved by combining pretraining skills.

Prefix-tuning may not learn a new task requiring a different attention pattern.

To check if prefix-tuning can learn a new task, we train a 1-layer, 1-head transformer to sort numbers into ascending order and then fine-tune it to sort in descending order. During training, the model sees random sequences of 10 digits from 0 to 7 followed by their ascending sorted order. The pretrained accuracy (fully matching the sorted sequence) is 91%. Full fine-tuning on the descending task leads to 85% test accuracy, hence full fine-tuning successfully learns the new task. However, prefix-tuning with a prefix size $n_S=1$ results in 0% accuracy, hence prefix-tuning fails to learn the new task at all.

The attention patterns in Figure 1 show why this is the case: the pretrained model learns to attend first to the smallest numbers and then to the larger ones. When fully fine-tuned, the attention patterns are reversed: they now first attend to the largest values. However, following Section 4, prefix-tuning cannot change the attention pattern over the input sequence and will still attend to the smallest values. Hence, prefix-tuning may indeed struggle to learn a new task requiring new attention patterns.

Prefix-tuning can elicit a skill from the pre-trained model.

The second part of our hypothesis was that prefix-tuning can elicit latent skills in the pretrained model. To test that, we pretrain a 1-layer, 4-head model with solutions sorted in ascending ($\nearrow$) or descending ($\searrow$) order, or adding one (+1) or two (+2) to each element of the input sequence. Each solution is shown with 25% probability. The model has no indication of what the task is, hence, it assigns equal probability to all tasks, as shown in the first row in Table 2. Full fine-tuning for each task naturally results in high accuracy. However, prefix-tuning ($n_S=1$) can also reach accuracy above 90% for all tasks. Compared to the previous case, prefix-tuning is more successful here because the pretrained model contains the attention mechanisms for solving the four tasks, as shown in Figure 2.

If all a prefix does is bias the attention layer activations, how can it steer the model to collapse its distribution onto one task? This is likely due to the attention block solving all tasks in parallel and placing their solutions in different subspaces of the residual stream (intermediate representation, Elhage et al., 2021). As the MLP needs to select one solution to generate, a further indicator on the selected task (or lack of selection thereof) should also be represented. The bias induced by the prefix then acts on this “selection subspace” to nudge the MLP to select the desired solution.

Table 1: A transformer pretrained on sorting in ascending order cannot be prefix-tuned to sort in descending order. 10 random seeds.

	Ascending	Descending
Pretrain on asc.	91±5%	0±0%
Full fine-tune on desc.	0±0%	85±5%
Prefix-tune on desc.	0±0%	0±0%

Table 2: A transformer pretrained on several tasks can be prefix-tuned for one of them. 10 random seeds.

Accuracy on:	$\nearrow$	$\searrow$	+1	+2
Pretrained	25±13%	25±12%	24±11%	22±7%
Prefix-tune on $\nearrow$	95±2%	0±0%	0±0%	0±0%
Prefix-tune on $\searrow$	0±0%	90±3%	1±1%	1±1%
Prefix-tune on +1	0±0%	1±3%	95±6%	0±1%
Prefix-tune on +2	0±0%	0±0%	1±2%	98±5%

⁷A similar hypothesis has also been proposed by Reynolds and McDonell (2021) for fine-tuning in general.

This can be clearly seen from the activations of the attention layer at the last input position (x_{n_x}): the position where the task selection happens as the first output element fully describes the task. Figure 3 shows plots of randomly selected dimensions of the residual stream with and without a prefix. The attention block activations of the pretrained model (without prefix) show no correlation with the output it is about to generate, demonstrating that the choice of completion is indeed not determined by the attention block. However, the prefix-tuned activations for the same inputs are clustered as a result of the prefix-induced bias. This indicates that the bias induced by the prefix may act as a “task selector” of the subspace of the residual stream specializing in the desired task.

Prefix-tuning can combine knowledge from pretraining tasks to solve new tasks. Prefix-tuning eliciting one type of completion learned in pretraining starts to explain its practical utility. Still, prefix-tuning seems to be successful also at tasks that the pretrained model has not seen. As we showed above, a model trained to sort in one order cannot be prefix-tuned to sort in the other. Then how is it possible for prefix-tuning to learn a new task? We posit that this can happen, as long as the “skill” required to solve the new task is a combination of “skills” the pretrained model has seen.

We test this by pretraining a 40-layer 4-head model with the same four tasks. We prefix-tune ($n_S=12$) for two new tasks: incrementing the ascending sorted sequence ($\nearrow+1$) and double histogram (mapping each element to the number of elements with the same value, e.g., $3,0,0,1 \mapsto 1,2,2,1$, $\mathbb{H}$). The pretrained model has not seen either task. Prefix-tuning results in 93% accuracy for $\nearrow+1$ which is a combination of the $\nearrow$ and $+1$ pretraining tasks and just 1% for the $\mathbb{H}$ task which requires different skills: finding other instances of the same token and counting. $\mathbb{H}$ is not a hard task: it requires 2 layers and 2 heads to be solved exactly (Weiss et al., 2021). Therefore, prefix-tuning is can indeed combine skills that the model has learned in order to solve a novel task but may not learn a completely new task requiring new skills.

Table 3: Prefix tuning can learn a new task requiring only pretraining skills ($\nearrow+1$) but cannot learn a completely new task ($\mathbb{H}$). Average accuracy over 3 seeds.

Accuracy on:	$\nearrow$	$\searrow$	$+1$	$+2$	$\nearrow+1$	$\mathbb{H}$
Pretrained	17%	23%	34%	25%	0%	0%
Prefix-tune on $\nearrow$	100%	0%	0%	0%	0%	0%
Prefix-tune on $\searrow$	0%	100%	0%	0%	0%	0%
Prefix-tune on $+1$	0%	0%	100%	0%	0%	0%
Prefix-tune on $+2$	0%	0%	0%	100%	0%	0%
Prefix-tune on $\nearrow+1$	0%	0%	0%	0%	93%	0%
Prefix-tune on $\mathbb{H}$	0%	0%	0%	0%	0%	1%

6 EFFECTS OF PREFIX-TUNING BEYOND THE SINGLE ATTENTION LAYER

Section 4 focused exclusively on a single attention layer. Still, even if a prefix only induces a bias on its output, this bias can exhibit complex behaviors via the subsequent MLPs and attention layers. This section shows how a prefix can change the attention pattern of the following attention layer but only in a linear fashion while full fine-tuning also has bilinear effects. Appendix C further argues that the representational capacity of prefix-tuning may be limited. Therefore, prefix-tuning appears to be less expressive than full fine-tuning, even with the same number of learnable parameters.

Prefix-tuning can change the attention, albeit the one of the next layer Let us examine how the prefix of one attention layer affects the following one. Assume no MLPs, residual connections or layer norms: the output $t_i^{(1)}$ of the first is the input $x_i^{(2)}$ of the second. The pretrained outputs are $t_i^{(1)} = \sum_{j=1}^p A_{ij}^{(1)} W_V^{(1)} x_j^{(1)}$, resulting in the second layer attention $\tilde{A}_{ij}^{(2)} = T/\sqrt{k} t_i^{(1)\top} H^{(2)} t_j^{(1)}$. Here $\tilde{A}_{ij}$ is the pre-softmax attention, i.e., $A_{ij} = \exp \tilde{A}_{ij} / \sum_{r=1}^p \exp \tilde{A}_{ir}$. For prefix-tuning we then have:

$$\begin{aligned}
 t_i^{\text{pt}(1)} &= A_{i0}^{\text{pt}(1)} W_V s_1^{(1)} + \sum_{j=1}^p A_{ij}^{\text{pt}(1)} W_V^{(1)} x_j^{(1)} \stackrel{(7)}{=} \underbrace{A_{i0}^{\text{pt}(1)}}_{\alpha_i} \underbrace{W_V s_1^{(1)}}_{\mu} + (1 - A_{i0}^{\text{pt}(1)}) t_i^{(1)}, \\
 \tilde{A}_{ij}^{\text{pt}(2)} &= \frac{T}{\sqrt{k}} t_i^{\text{pt}(1)\top} H^{(2)} t_j^{\text{pt}(1)}, \\
 &= \frac{T}{\sqrt{k}} (\underbrace{\alpha_i \alpha_j}_{\text{constant}} \underbrace{\mu^\top H^{(2)} \mu}_{\text{depends only on } t_i^{(1)}} + \underbrace{\alpha_j (1 - \alpha_i)}_{\text{depends only on } t_i^{(1)}} \underbrace{t_i^{(1)\top} H^{(2)} \mu}_{\text{depends only on } t_j^{(1)}} + \underbrace{\alpha_i (1 - \alpha_j)}_{\text{depends only on } t_j^{(1)}} \underbrace{\mu^\top H^{(2)} t_j^{(1)}}_{\text{depends only on } t_j^{(1)}} + (1 - \alpha_i)(1 - \alpha_j) \underbrace{t_i^{(1)\top} H^{(2)} t_j^{(1)}}_{\text{pretrained attention } \tilde{A}_{ij}^{(2)}}).
 \end{aligned}$$

The presence of μ shows that the prefix of layer 1 can change the attention pattern of the following layer. This change is content-specific: the second and the third terms depend on the inputs, hence a simple bias can affect the attention when passed through MLPs and further attention blocks. Compare with Equation (6), which showed a prefix cannot change the attention of the same layer. Still,

even considering this cross-layer effect, prefix-tuning is more limited in its expressiveness than full fine-tuning. While the second and the third terms are input-dependent, each depends on one input position only. The prefix does not change the bilinear dependency on both the query and key. This is something that the full fine-tuning can achieve: $\tilde{A}_{ij}^{(2)} = \tau/\sqrt{k} \mathbf{t}_i^{(1)\top} (\mathbf{H}^{(2)} + \Delta\mathbf{H}^{(2)}) \mathbf{t}_j^{(1)}$.

Even if prefix-tuning could be a universal approximator, it would not be parameter-efficient.

Prefix-tuning appears to be less parameter-efficient than other comparable approaches. We designed an experiment to this end. Our pretrained model in Section 5 failed to learn the double histogram task ($\mathbb{H}$). A rank-1 Low Rank Adaptation (LoRA, Hu et al., 2021) applied only to the MLPs in a 4-layer 4-head model pretrained in the exact same way results in 92% accuracy on the $\mathbb{H}$ task. The number of parameters for the LoRA fine-tuning is exactly the same as for a prefix of size 12. However, as can be expected from the results in Section 5, training this prefix results in 0% accuracy. Hence, prefix-tuning fails at a task that LoRA with the same number of parameters can learn.

In conclusion, while prefix-tuning, prompting, soft prompting and in-context learning have complex effects in deeper models, the interaction of the learnable parameters and the model inputs likely still results in very limited expressiveness. In particular, we demonstrated that LoRA can be used to learn a completely new task while prefix-tuning with the exact same number of parameters fails to.

7 DISCUSSION AND RELATED WORKS

Understanding fine-tuning and prefix-tuning. Prior works show that prefixes have low intrinsic dimension allowing transfer to similar tasks and initialization of prefixes for new tasks (Qin et al., 2021; Su et al., 2022; Zhong et al., 2022; Wang et al., 2022b; Zheng et al., 2023). In this work, we offered theoretical insights into their results: this subspace is the span of the prefix-induced bias. Another line of work shows that skills can be localized in the parameter space of pretrained models (Wang et al., 2022a; Panigrahi et al., 2023). Here, we showed that it is also possible to identify subspaces of the residual stream corresponding to individual tasks and select them via prefix-tuning.

Prompting and in-context learning. Prompting and in-context learning are a special case of prefix-tuning. Therefore, the limitations and mechanisms discussed in this work apply to prompting as well: prompts cannot change the distribution of attention of the first attention layer over the content following it and can only induce a bias on the output of this layer (Section 4). Even considering the cross-layer effects, a prompt is strictly less expressive than full fine-tuning (Section 6) and prompting is unlikely to enable the model to solve a completely new task. Our theory thus explains why Kossen et al. (2023) observed that in-context examples cannot overcome pre-training skills.

While context-based fine-tuning approaches may not learn *arbitrary* new tasks, as shown in Section 5, they can leverage pre-trained skills. Wies et al. (2023) have PAC-learnability results that also show that when pretraining is on a mixture tasks, they can be efficiently learned via in-context learning. Moreover, transformers can learn linear models in-context by mimicking gradient descent (Von Oswald et al., 2023) or approximating matrix inversion (Akyürek et al., 2022). This is consistent with our theory: the prediction updates are enacted as biases in the attention block activations. Hence, despite the limitations discussed in this work, context-based methods can result in powerful fine-tuning if the pretrained model has “transferable skills” such as algorithmic fundamentals. Still, in-context learning will likely fail for non-algorithmic tasks, e.g., translating to a language that the model has never seen before, even if large number of translation pairs are provided in-context.

Implications for catastrophic forgetting and model alignment. The lack of expressiveness of context-based fine-tuning can be a feature: desirable properties will be maintained. Full fine-tuning can result in catastrophic forgetting (He et al., 2021b; Luo et al., 2023; Mukhoti et al., 2023). Our theory shows that context-based methods won’t lose pretrained skills. Model alignment poses the reverse problem: ensuring that the model cannot pick up undesirable skills during fine-tuning. Our results show that prompting and prefix-tuning might be unable to steer the model towards new adversarial behaviors. Hence, the recent successes in adversarial prompting (Zou et al., 2023) indicate that current model alignment methods just mask the undesirable skills rather than removing them.

Implications for model interpretability. An open question for language model interpretability is whether attention is sufficient for explainability (Jain and Wallace, 2019; Wiegrefe and Pinter, 2019). Section 5 points toward the negative: by interfering in the *output* of the attention layer with the bias induced by a prefix, we can change the behavior of the model, without changing its attention.